**SIMATS ENGINEERING**
**CO3 - ASSESSMENT TOOL - 2: Case Study**

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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|---|---|
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**COURSE OUTCOME COVERED**
**CO3:** *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

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**Code of Conduct**
I ___E.B.__Ashwini/192511422________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: _Ashwini____________________

**Assessment Tasks - Case Study**

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.
2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.
3. **Case Study 3: Online Food Ordering Platform**

   A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

**Case Study Rubric**

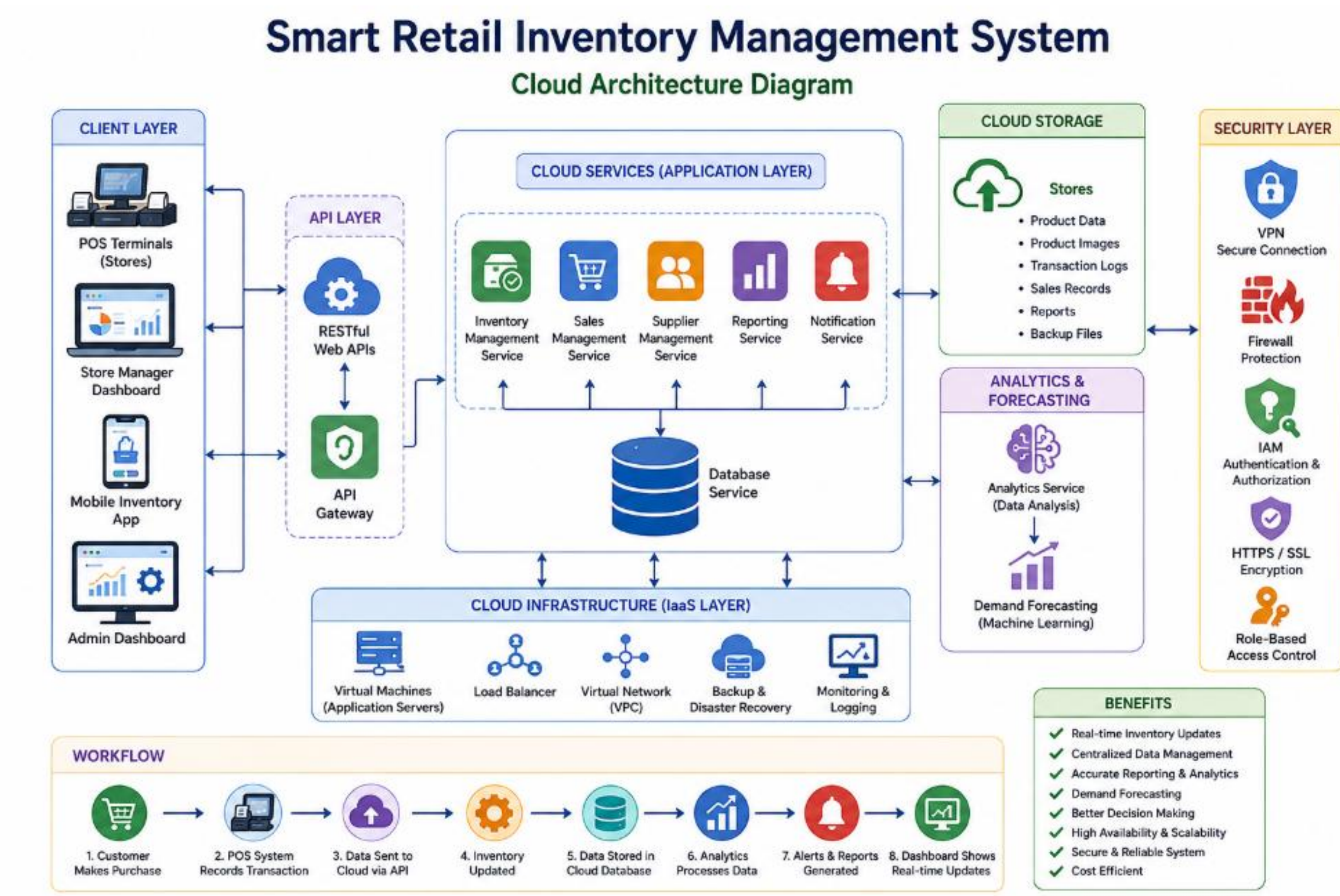
| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Understanding of Case | 25% | Clearly identifies all technical and business requirements | Identifies most requirements | Basic understanding only | Poor understanding of case |
| Application of Concepts | 25% | Applies suitable cloud concepts and tools effectively | Good application with minor issues | Partial application | Weak or incorrect application |
| Analysis | 20% | Strong analysis with clear trade-off reasoning | Reasonable analysis | Limited analysis | Minimal analysis |
| Solution Design | 20% | Solution is feasible, practical, and well justified | Reasonable solution with minor gaps | Basic solution | Unclear solution |
| Clarity | 10% | Well-structured and easy to follow | Mostly clear | Somewhat unclear | Poorly organized |

**Case Study 1: Smart Retail Inventory Management System**

**A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.**

Retail businesses need an efficient way to manage their inventory across multiple stores. Traditional inventory systems often store data locally, making it difficult to keep stock information updated in real time. This can result in problems such as overstocking, stock shortages, delayed updates, and inaccurate reports. As the business grows, managing inventory manually becomes more complex and expensive.

## Cloud Architecture of Smart Retail Inventory Management System




To solve these challenges, the retail chain plans to implement a cloud-based inventory management system. In this system, all store branches are connected to a centralized cloud platform. Point of Sale (POS) systems, inventory dashboards, and mobile applications communicate with cloud services through Web APIs. Cloud storage securely stores inventory data, sales records, and customer transaction logs. Cloud platform services process this data to generate reports, analyze sales trends, and forecast future product demand. Security measures such as VPNs, firewalls, API gateways, and encryption ensure that business data remains safe.

This solution allows the company to monitor inventory in real time, improve business efficiency, reduce operational costs, and provide better customer service.

## Understanding of the Case

- The case study describes a retail company that wants to modernize its inventory management by using cloud computing technologies. Instead of maintaining separate databases in each store, all inventory information is stored in a centralized cloud environment. Whenever a product is sold, the POS system immediately sends the transaction details to the cloud through a RESTful API. The inventory database is updated automatically, and managers can instantly view the latest stock information through dashboards.
- The cloud platform also collects sales data from every branch and uses analytics services to identify customer buying patterns. Based on this information, the system predicts future demand, helping the company maintain the right amount of stock and avoid unnecessary losses.
- The proposed system focuses on improving efficiency, scalability, security, and business decision-making.

## Business Requirements

The organization expects the cloud-based inventory system to achieve the following business objectives:

- Maintain a centralized inventory database for all stores.
- Update stock levels immediately after every sale.
- Reduce manual inventory management.
- Improve customer satisfaction by ensuring product availability.
- Generate automated sales and inventory reports.
- Forecast product demand using analytics.
- Reduce operational and maintenance costs.

- Support business growth by allowing easy expansion to new store locations.

## Technical Requirements

To meet the business objectives, the following technical components are required:

## Cloud Storage

Cloud storage is used to securely store:

- Product details
- Inventory information
- Sales transaction records
- Customer purchase history
- Supplier information
- Business reports
- Analytics datasets

## RESTful Web APIs

REST APIs allow communication between:

- POS systems and cloud servers
- Inventory dashboards and databases
- Mobile applications and cloud services

## Platform Services (PaaS)

Cloud platform services provide:

- Data analytics
- Machine learning for demand forecasting
- Notification services
- Monitoring and logging

- Database management

## Infrastructure (IaaS)

Infrastructure services include:

- Virtual Machines
- Virtual Networks
- Load Balancers
- Cloud Databases
- Backup Servers

## Cloud Architecture

The proposed cloud architecture consists of several layers that work together to provide a reliable and scalable inventory management system.

## Client Layer

The client layer includes:

- Point of Sale (POS) terminals
- Store manager dashboards
- Mobile inventory applications
- Administrator portal

Users interact with the system through these devices.

## API Layer

The API layer receives requests from clients and sends responses back after processing the information.

Examples of API operations include:

- Viewing product details
- Updating inventory

- Recording sales
- Generating reports

## Application Layer

This layer contains the business logic of the system.

Its responsibilities include:

- Inventory management
- Sales processing
- Report generation
- Customer management
- Supplier management

## Data Layer

The data layer stores all information in cloud databases and cloud storage.

## Working Process :

The cloud inventory management system operates through the following sequence:

**Step 1:** A customer purchases a product at the retail store.

**Step 2:** The POS system records the transaction.

**Step 3:** The transaction details are sent to the cloud server through a RESTful API.

**Step 4:** The API Gateway verifies the authenticity of the request before allowing access.

**Step 5:** The application server processes the transaction.

**Step 6:** The inventory database automatically decreases the available stock.

**Step 7:** The sales transaction is saved in cloud storage.

**Step 8:** Analytics services update business reports and identify sales trends.

**Step 9:** If stock falls below the minimum level, the system automatically sends a notification to the store manager.

**Step 10:** Managers view updated inventory information through cloud dashboards.

## Cloud Services Used

## Infrastructure as a Service (IaaS)

Infrastructure services provide the computing resources required to run the application.

Examples include:

- Virtual Machines
- Load Balancers
- Virtual Networks
- Storage Services

Benefits:

- Easy scalability
- Reduced hardware investment
- High availability

## Platform as a Service (PaaS)

Platform services simplify application development and deployment.

Examples include:

- Database services
- Analytics platforms
- Machine learning services
- Monitoring tools

Benefits:

- Faster application development
- Automatic software updates
- Reduced maintenance

## Software as a Service (SaaS)

End users access inventory dashboards and reporting applications through web browsers without installing additional software.

Benefits:

- Easy accessibility
- Automatic updates
- Simple user management

## Cloud Storage

Cloud storage plays an important role in this solution.

It stores:

- Product information
- Product images
- Sales invoices
- Transaction logs
- Inventory reports
- Analytics data
- Backup files

## Benefits of Cloud Storage

- Data can be accessed from anywhere.
- Automatic backup reduces the risk of data loss.

- Storage capacity can be increased whenever required.
- Multiple copies of data improve reliability.
- High-speed access improves application performance.

## Web APIs

RESTful APIs act as the communication bridge between client applications and cloud services.

Common APIs include:

- View Products
- Add Product
- Update Product
- Delete Product
- Record Sales
- Check Inventory
- Generate Reports

Benefits:

- Faster communication
- Platform independence
- Easy integration with mobile and web applications
- Secure data exchange

## Platform Services

Cloud platform services provide intelligent features such as:

## Data Analytics

Analyzes sales data to identify:

- Best-selling products

- Seasonal demand
- Customer buying behavior

## Demand Forecasting

Machine learning algorithms estimate future product demand based on historical sales data.

### Notification Service

Automatically sends alerts when:

- Inventory becomes low
- New orders arrive
- System errors occur

## Security Measures

Security is one of the most important aspects of the proposed system.

## Virtual Private Network (VPN)

Provides secure communication between store branches and cloud servers.

## Firewall

Prevents unauthorized users from accessing cloud resources.

## API Gateway

Protects APIs by validating every request before it reaches the application.

## HTTPS

Encrypts all data transmitted between clients and cloud servers.

## Identity and Access Management (IAM)

Allows only authorized employees to access the system.

## Role-Based Access Control (RBAC)

Different users have different permissions.

| User | Access Level |
| --- | --- |
| Administrator | Full Access |
| Store Manager | Inventory and Reports |
| Cashier | Billing Only |
| Warehouse Staff | Inventory Updates |

## Advantages of the Proposed System

The cloud-based inventory management system provides several advantages:

- Real-time inventory updates.
- Centralized management across all stores.
- Improved business decision-making using analytics.
- Automatic demand forecasting.
- Reduced operational costs.
- Better customer satisfaction.
- Secure storage of business information.
- Automatic backup and disaster recovery.
- Easy expansion when new stores are opened.
- High availability with minimal downtime.

## Challenges

Although the system offers many benefits, it also has a few challenges:

- Continuous internet connectivity is required.

- Initial migration to the cloud may be expensive.
- Employees need training to use the new system.
- Cloud security policies must be managed properly.
- Dependence on the cloud service provider.

## Analysis and Trade-Offs

The cloud solution provides significant improvements compared to traditional inventory management systems.

| Cloud-Based System | Traditional System |
|---|---|
| Real-time inventory updates | Manual updates |
| Centralized database | Separate databases |
| Automatic backups | Manual backups |
| Easy scalability | Limited scalability |
| Advanced analytics | Basic reporting |
| High availability | Risk of server failure |

Although cloud computing introduces internet dependency and ongoing subscription costs, these disadvantages are outweighed by the benefits of scalability, reliability, automation, and better business insights.

## Proposed Solution

The proposed Smart Retail Inventory Management System should be implemented using:

- Cloud Storage for centralized data management.
- RESTful Web APIs for communication between clients and servers.
- Virtual Machines and Load Balancers for hosting.
- Platform services for analytics and demand forecasting.

- API Gateway, VPN, HTTPS, and Firewalls for security.
- Cloud monitoring tools for performance tracking.

This architecture ensures that the system remains scalable, secure, reliable, and capable of supporting future business growth.

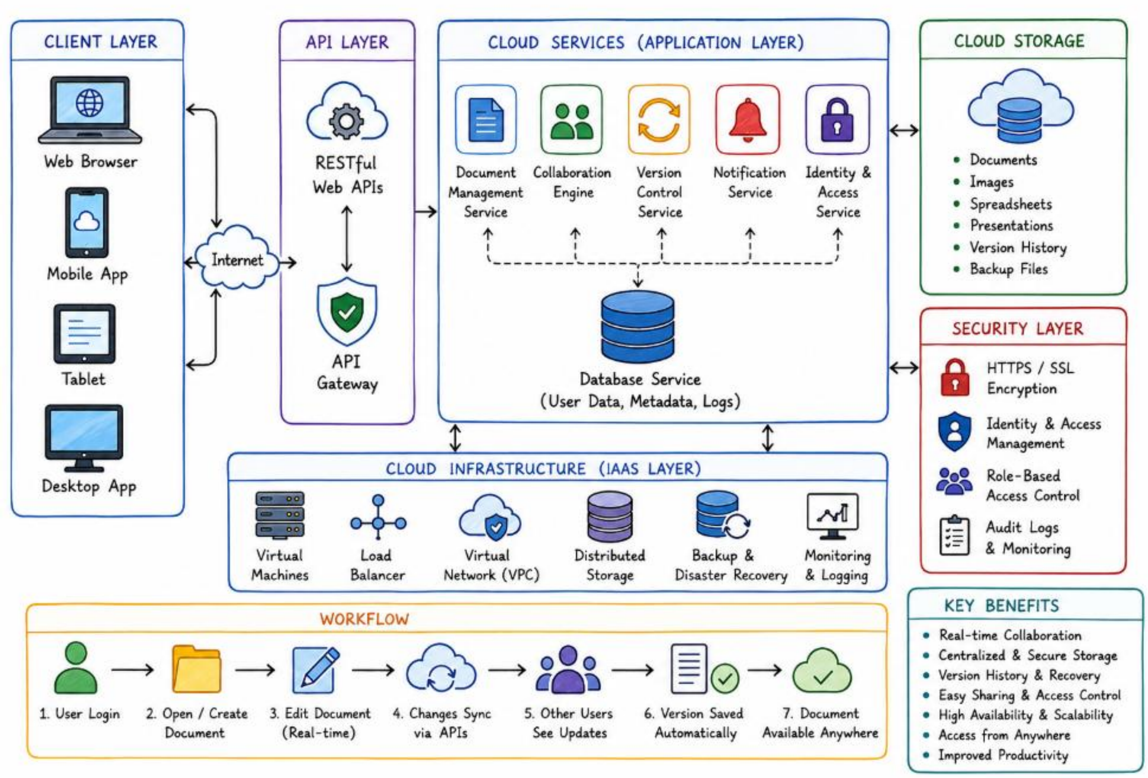
**Case Study 2: Cloud-Based Document Collaboration System**

**An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.**

In today's digital world, organizations need fast and efficient ways to collaborate on documents, especially when employees work from different locations. Traditional document management systems often require users to save multiple copies of the same file, making it difficult to track changes and collaborate effectively. This can lead to duplicate files, version conflicts, and delays in communication.

To overcome these challenges, the enterprise plans to implement a Cloud-Based Document Collaboration System, similar to Google Docs. The system enables multiple users to create, edit, and share documents simultaneously using web browsers or mobile devices. All documents are securely stored in distributed cloud storage, while RESTful Web APIs synchronize changes across all connected devices in real time. The cloud platform ensures high availability, scalability, and reliable performance, while advanced security features such as encryption, identity management, and access permissions protect business information.

## Architecture of the Cloud-Based Document Collaboration System



The proposed solution improves teamwork, increases productivity, and provides secure access to documents from anywhere in the world.

## Understanding of the Case

The enterprise wants to build a cloud-based document collaboration platform where employees can work together on the same document in real time. Instead of sending documents through emails or storing multiple copies on local computers, all documents are maintained in a centralized cloud environment.

Whenever a user edits a document, the changes are instantly synchronized with the cloud server through RESTful Web APIs. Every authorized user viewing the document immediately sees the latest updates without refreshing the page. The system also maintains version history, allowing users to restore previous versions if needed.

To ensure smooth collaboration across different countries and devices, the cloud infrastructure uses distributed storage, load balancing, and global networking services. Security mechanisms such as HTTPS, encryption, identity management, and role-based access control protect confidential documents from unauthorized access.

Overall, the system provides a reliable, secure, and efficient environment for document creation, editing, and collaboration.

## Business Requirements

The organization expects the cloud-based collaboration system to achieve the following objectives:

- Allow multiple users to edit documents simultaneously.
- Provide real-time synchronization across all devices.
- Store documents securely in the cloud.
- Maintain document version history.
- Enable easy document sharing within the organization.
- Improve employee productivity and collaboration.
- Reduce document duplication.
- Ensure secure access to sensitive business documents.

- Support remote work from any location.
- Provide high availability and reliability.

## Technical Requirements

To successfully implement the solution, the following technical components are required.

### Cloud Storage

Cloud storage is used to store:

- Documents
- Images
- Spreadsheets
- Presentation files
- User profiles
- Version history
- Backup files
- Access permissions

## RESTful Web APIs

REST APIs enable communication between:

- Web browsers
- Mobile applications
- Cloud servers
- Database services

Common API operations include:

- Create document
- Open document

- Edit document
- Save document
- Share document
- Delete document
- Restore previous version

## Platform Services (PaaS)

Platform services provide:

- Document synchronization
- Version control
- Authentication services
- Notification services
- Database management
- Monitoring and logging

## Infrastructure (IaaS)

Infrastructure services include:

- Virtual Machines
- Load Balancers
- Virtual Networks
- Cloud Databases
- Distributed Storage
- Backup Servers

## Cloud Architecture

The cloud-based collaboration system consists of several layers.

**Client Layer**

Users access the application using:

- Web Browser
- Mobile App
- Tablet
- Desktop Application

Employees can create, edit, and share documents from any device connected to the internet.

**API Layer**

The API layer handles communication between client applications and cloud services.

Functions include:

- User authentication
- Document synchronization
- File upload
- File download
- Version management

**Application Layer**

The application layer manages:

- Document editing
- Collaboration engine
- Version control
- User management
- Sharing permissions
- Notifications

**Data Layer**

The data layer stores:

- Documents
- Images
- User accounts
- Version history
- Backup files
- Audit logs

All information is stored securely in distributed cloud storage.

## Working Process

The document collaboration system follows the steps below.

**Step 1:**

The user logs into the system using secure authentication.

**Step 2:**

The user opens an existing document or creates a new one.

**Step 3:**

The browser sends the request to the cloud server through a RESTful Web API.

**Step 4:**

The application verifies the user's identity and access permissions.

**Step 5:**

The requested document is retrieved from cloud storage.

**Step 6:**

The user edits the document.

**Step 7:**

Every modification is automatically synchronized with the cloud server.

**Step 8:**

Other authorized users immediately see the latest changes.

**Step 9:**

The version control service saves every update.

**Step 10:**

If required, users can restore previous versions of the document.

## Cloud Service Models

### Infrastructure as a Service (IaaS)

Provides computing resources such as:

- Virtual Machines
- Virtual Networks
- Storage
- Load Balancers

  Benefits:

- Easy scalability
- High availability
- Reduced hardware cost

### Platform as a Service (PaaS)

Provides development and application services such as:

- Collaboration engine
- Database services

- Authentication
- Monitoring
- Version control

Benefits:

- Faster development
- Automatic software updates
- Easy deployment

**Software as a Service (SaaS)**

End users access the document collaboration platform directly through a web browser without installing additional software.

Benefits:

- Easy access
- Automatic updates
- Simple maintenance

## Cloud Storage

Cloud storage stores all business documents securely.

It contains:

- Word documents
- PDF files
- Excel sheets
- Images
- Presentations
- Version history

- Backup files

**Benefits**

- Secure storage
- Automatic backup
- High availability
- Global accessibility
- Easy scalability
- Fast data recovery

## RESTful Web APIs

REST APIs enable communication between client applications and cloud services.

Examples include:

- Create Document
- Open Document
- Save Document
- Edit Document
- Share Document
- Delete Document
- Restore Version

**Benefits**

- Real-time synchronization
- Faster communication
- Platform independence
- Easy integration

- Secure data transfer

## Platform Services

The platform services provide intelligent features.

## Collaboration Engine

Allows multiple users to edit the same document simultaneously without conflicts.

## Version Control

Maintains previous versions of documents.

Users can restore older versions whenever necessary.

## Notification Service

Automatically informs users about:

- Shared documents
- Comments
- Editing activities
- Access requests

### Identity Management

Verifies user identity before granting access.

## Network Infrastructure

The cloud infrastructure includes:

- Distributed Cloud Storage
- Load Balancer
- Content Delivery Network (CDN)
- Virtual Private Cloud
- DNS

- Global Data Centers

These components ensure low latency, fast document access, and high availability worldwide.

## Advantages of the Proposed System

The cloud-based document collaboration system offers several advantages:

- Real-time collaboration
- Easy document sharing
- Automatic synchronization
- Centralized document management
- High availability
- Improved productivity
- Secure access
- Automatic backup
- Easy scalability
- Reduced storage cost
- Global accessibility
- Version history management

## Challenges

Although the proposed system has many benefits, it also has some limitations.

- Requires a stable internet connection.
- Large files may take longer to upload.
- Cloud storage subscription costs.
- Data privacy regulations must be followed.
- Synchronization conflicts may occur during poor network conditions.

## Analysis and Trade-Offs

| Cloud-Based Collaboration | Traditional Document Sharing |
|---|---|
| Real-time editing | Single-user editing |
| Automatic synchronization | Manual file sharing |
| Centralized cloud storage | Local storage |
| Version history | Multiple duplicate files |
| Easy remote access | Limited remote access |
| Automatic backup | Manual backup |
| High availability | Depends on local servers |
| Secure access control | Basic file permissions |

Although cloud collaboration depends on internet connectivity and requires cloud service costs, these disadvantages are outweighed by improved teamwork, security, scalability, and productivity.

## Proposed Solution

The enterprise should implement the system using:

- Distributed Cloud Storage for secure document storage.
- RESTful Web APIs for real-time synchronization.
- Collaboration Engine for simultaneous editing.
- Version Control Service to maintain document history.
- Identity and Access Management (IAM) for authentication.
- Role-Based Access Control (RBAC) for user permissions.
- HTTPS and SSL/TLS encryption for secure communication.
- Load Balancers and Content Delivery Networks (CDN) for high availability and low latency.

- Cloud monitoring services for performance tracking and fault detection.

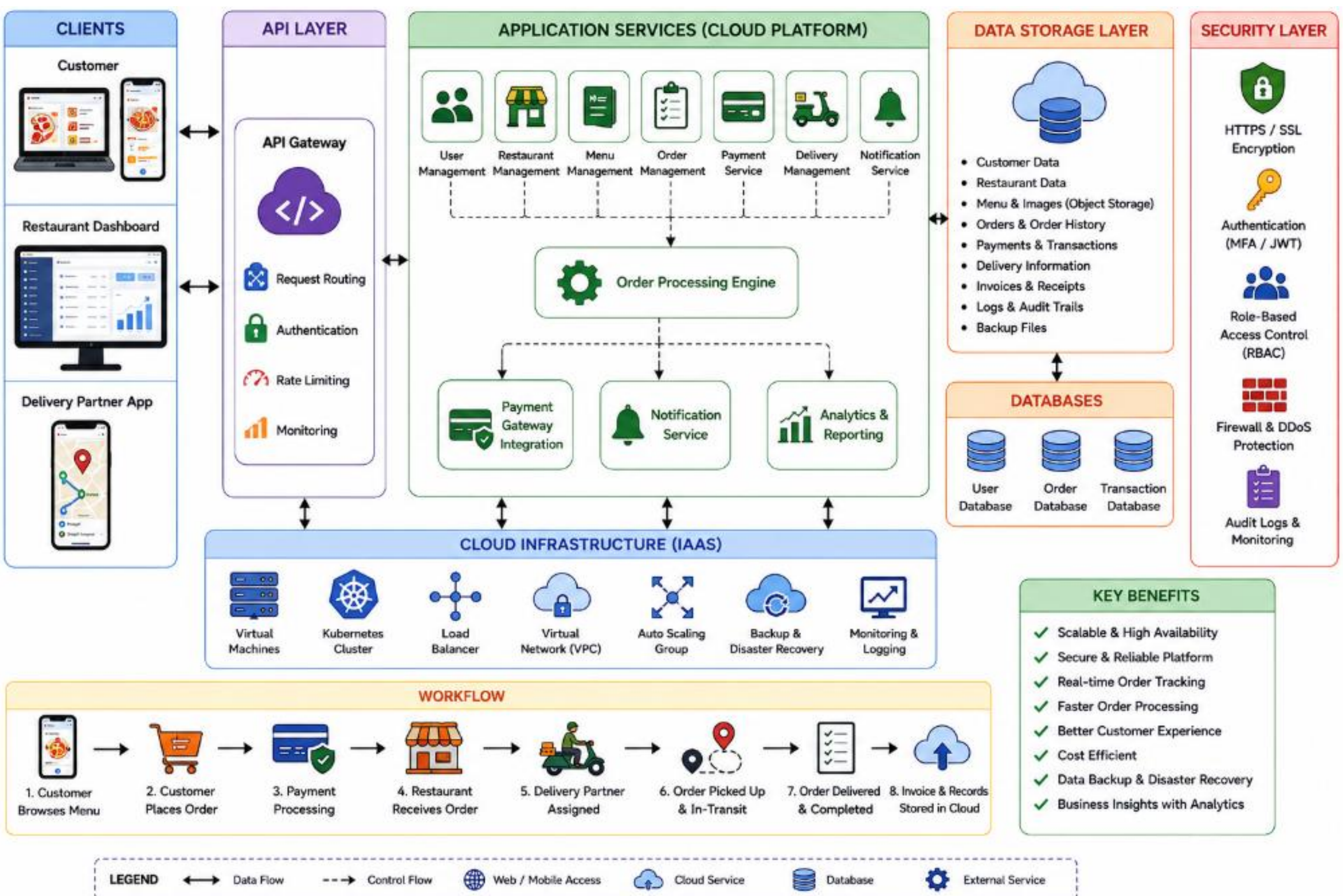
This architecture provides a secure, scalable, and reliable environment for document collaboration while supporting users across different locations and devices.

**Case Study 3: Online Food Ordering Platform**

**A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients.Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.**

The food delivery industry has experienced rapid growth due to increasing internet access and the widespread use of smartphones. Customers expect a fast, secure, and convenient way to order food online without visiting restaurants. To meet these expectations, the startup plans to develop a Cloud-Based Online Food Ordering Platform that can be accessed through web browsers and mobile applications.

## Architecture of the Cloud-Based Online Food Ordering Platform




The proposed system uses cloud computing technologies to provide a scalable, reliable, and secure platform for customers, restaurants, delivery partners, and administrators. Customers browse restaurant menus, place orders, make online payments, and track deliveries in real time. RESTful Web APIs connect the frontend applications with backend cloud services. Cloud storage securely stores menu images, invoices, customer information, and order history. The application is deployed using virtual machines and managed Kubernetes clusters to ensure high availability and automatic scalability during peak hours. Security is maintained through HTTPS, authentication tokens, encryption, and role-based access control.

This cloud-based solution improves customer experience, simplifies restaurant operations, and enables the startup to expand its services efficiently.

## Understanding of the Case

The case study describes an online food ordering system that allows customers to order food from different restaurants using a web browser or mobile application. Instead of managing separate systems for orders, payments, and deliveries, the platform integrates all services into a centralized cloud environment.

Whenever a customer places an order, the request is sent through RESTful Web APIs to the cloud server. The system processes the order, updates the restaurant dashboard, sends payment information to the payment gateway, and notifies the assigned delivery partner. Cloud storage securely maintains customer details, menu images, invoices, and order history.

The cloud infrastructure uses virtual machines, Kubernetes, and load balancing to handle large numbers of users simultaneously. Security features such as HTTPS, JSON Web Tokens (JWT), encryption, and role-based access control ensure that customer and business information remains protected.

Overall, the platform provides a secure, scalable, and reliable solution for modern online food delivery services.

## Business Requirements

The startup expects the cloud-based food ordering system to achieve the following objectives:

- Allow customers to browse restaurant menus online.
- Enable customers to place food orders quickly.
- Support secure online payment methods.
- Provide real-time order tracking.
- Allow restaurants to manage incoming orders efficiently.

- Notify delivery partners about assigned orders.
- Improve customer satisfaction through faster service.
- Reduce manual order processing.
- Support business growth by handling increasing numbers of users.
- Ensure secure storage of customer and transaction data.

## Technical Requirements

To successfully implement the platform, the following technical components are required.

## Cloud Storage

Cloud storage securely stores:

- Menu images
- Customer information
- Restaurant details
- Order history
- Digital invoices
- Payment records
- Delivery information
- Application logs
- Backup files

## RESTful Web APIs

REST APIs provide communication between:

- Web browsers
- Mobile applications
- Cloud servers

- Restaurant dashboards
- Payment gateway
- Delivery management system
- User Registration
- User Login
- View Menu
- Place Order
- Make Payment
- Track Order
- Cancel Order
- Update Delivery Status

## Platform Services (PaaS)

Platform services provide:

- Order processing
- Notification services
- Database management
- Monitoring and logging
- Payment integration
- Analytics dashboard

## Infrastructure (IaaS)

Infrastructure services include:

- Virtual Machines
- Kubernetes Cluster

- Virtual Network
- Cloud Database
- Load Balancer
- Backup Server

## Cloud Architecture

The cloud-based food ordering platform consists of multiple layers that work together to deliver a smooth and reliable user experience.

**Client Layer**

Users access the application using:

- Web Browser
- Android Application
- iOS Application

Customers can browse menus, place orders, and track deliveries from any internet-connected device.

**API Layer**

The API layer receives requests from users and securely forwards them to the backend services.

Functions include:

- User authentication
- Menu retrieval
- Order processing
- Payment processing
- Delivery tracking

**Application Layer**

The application layer manages:

- Customer accounts
- Restaurant management
- Order management
- Payment processing
- Delivery management
- Notifications

**Data Layer**

The data layer stores:

- Customer profiles
- Restaurant information
- Orders
- Payments
- Delivery records
- Menu images
- Invoices

All data is securely stored in cloud databases and cloud object storage.

## Working Process

The online food ordering platform works as follows.

**Step 1**

The customer logs into the application using secure authentication.

**Step 2**

The customer searches for a restaurant and views the available menu.

**Step 3**

The customer selects food items and places an order.

**Step 4**

The order request is sent to the cloud server through a RESTful Web API.

**Step 5**

The application verifies the user's identity and processes the order.

**Step 6**

The restaurant receives the order notification and begins preparing the food.

**Step 7**

The customer completes payment through the secure payment gateway.

**Step 8**

The delivery partner receives the delivery request.

**Step 9**

The customer tracks the delivery status in real time.

**Step 10**

After successful delivery, the invoice and order details are securely stored in cloud storage.

## Cloud Service Models

### Infrastructure as a Service (IaaS)

Infrastructure services provide:

- Virtual Machines
- Virtual Networks
- Storage
- Load Balancers

**Benefits**

- Flexible computing resources
- High availability
- Easy scalability
- Reduced hardware costs

**Platform as a Service (PaaS)**

Platform services provide:

- Application hosting
- Database services
- Notification services
- Monitoring
- Analytics

**Benefits**

- Faster application development
- Automatic updates
- Easy deployment
- Reduced maintenance

**Software as a Service (SaaS)**

Customers access the food ordering platform through web browsers and mobile applications without installing complex software.

**Benefits**

- Easy accessibility
- Automatic updates
- Better user experience

## Cloud Storage

Cloud storage plays an important role in the system.

It stores:

- Restaurant menu images
- Customer profiles
- Order history
- Payment receipts
- Digital invoices
- Delivery records
- Backup files

**Benefits**

- Secure storage
- Automatic backup
- High availability
- Global accessibility
- Easy scalability
- Fast recovery

## RESTful Web APIs

REST APIs connect frontend applications with backend services.

Common APIs include:

- Register User
- Login User
- View Restaurants

- View Menu
- Add to Cart
- Place Order
- Process Payment
- Track Order
- View Order History

**Benefits**

- Fast communication
- Platform independence
- Secure data exchange
- Easy integration
- Real-time updates

## Platform Services

Platform services improve the overall performance of the system.

## Order Management Service

Processes customer orders and updates restaurants instantly.

## Payment Gateway

Handles secure online payments using digital payment methods.

**Notification Service**

Sends notifications to:

- Customers
- Restaurants
- Delivery partners

- Order confirmation
- Payment confirmation
- Delivery updates

**Analytics Service**

Analyzes:

- Popular food items
- Customer ordering patterns
- Restaurant performance
- Peak ordering hours

## Network Infrastructure

The cloud infrastructure includes:

- Virtual Private Cloud (VPC)
- Load Balancer
- Kubernetes Cluster
- Virtual Machines
- API Gateway
- DNS
- Content Delivery Network (CDN)

These components ensure low response time, high availability, and reliable service during peak traffic.

## Security Measures

Security is one of the most important aspects of the platform.

**HTTPS**

Encrypts communication between users and cloud servers.

**Authentication Tokens (JWT)**

Verifies user identity before allowing access to the system.

**Role-Based Access Control (RBAC)**

Different users have different permissions.

| User | Permission |
|---|---|
| Administrator | Full System Access |
| Restaurant Owner | Manage Menu and Orders |
| Delivery Partner | View Assigned Deliveries |
| Customer | Place Orders and Track Delivery |

**API Gateway**

Protects APIs by:

- Authenticating requests
- Filtering malicious traffic
- Monitoring API usage

## Advantages of the Proposed System

The cloud-based food ordering platform offers several advantages:

- Easy online food ordering
- Real-time order tracking
- Secure online payments
- Fast order processing
- Automatic scalability during peak hours
- High system availability
- Secure cloud storage

- Better customer experience
- Improved restaurant management
- Easy business expansion
- Automatic backup and recovery

## Challenges

Although the platform provides many benefits, it also faces some challenges.

- Requires a stable internet connection.
- Cloud infrastructure costs may increase as users grow.
- Managing Kubernetes clusters requires technical expertise.
- Security must be continuously monitored.
- High traffic during festivals or weekends may require additional cloud resources.

## Proposed Solution

The startup should implement the Online Food Ordering Platform using:

- Cloud Storage for menu images, invoices, and customer data.
- RESTful Web APIs for communication between clients and backend services.
- Managed Kubernetes for automatic scaling and high availability.
- Virtual Machines for hosting backend applications.
- Cloud Databases for storing customer, restaurant, and order information.
- Payment Gateway integration for secure transactions.
- API Gateway, HTTPS, JWT authentication, firewalls, and Role-Based Access Control (RBAC) for security.
- Cloud monitoring and logging services to track application performance and quickly identify issues.

This architecture ensures that the platform remains secure, scalable, and capable of supporting a growing customer base.